Anindya Mondal

Researcher and Engineer

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Career Highlights

- **7 peer-reviewed publications:** in **ACM TOG (SIGGRAPH Asia)**, **AAAI**, and **ICCV-W**, plus ICASSP and EUSIPCO, spanning vision, graphics, and multimodal learning.
- **4+ years of research experience:** Research Scientist/Engineer internships across academia and industry (incl. *Adobe Research*), plus doctoral research on generative and multimodal AI, including RL post-training (GRPO) and reward modeling of vision–language models.
- **4 years of teaching and supervision experience:** Supervised 10+ MSc students in their Master thesis at the University of Surrey and assisted teaching 2 courses in advanced CV/ML.

Research Focus

a) Multimodal Foundation Models (VLM, CLIP, SAM, Diffusion) - b) Controllable & Interactive Image/Video Generation (Diffusion, Agentic) - c) RL Post-Training & Alignment (GRPO, DPO, RLHF, Reward Modeling) - d) Efficient & Training-Free Adaptation of Foundation Models

Professional Experience

Research Intern, *(Adobe Research)* **Bengaluru, India** *06/2026 - present*

- Building rubric-aware VLM judges (reward models) to evaluate generated content across semantics, spatial structure, and object relationships.
- Designing criterion-level reward signals for prompt fidelity, edit correctness, layout preservation, and artifact severity.
- Refining the generation → understanding → feedback (reward) loop for aligning generative models.

Doctoral Researcher, *(University of Surrey, CVSSP)* **Guildford, UK** *10/2022 - present*

- Built **ABACUS** (ACM TOG), a unified generation/understanding VLM aligned with a cycle-consistent **GRPO (RL post-training)** objective for count-accurate, controllable image synthesis.
- Proposed **OmniCount** (AAAI), cutting counting error by ~**50%** on benchmark datasets, and released OmniCount-191 (30K images, 191 categories).
- Developed **CountLoop**, a training-free agent loop with an MLLM planner–critic for accurate 30-200 instance generation.
- Created **MSQNet** (ICCVW), an actor-agnostic action recognition model using multimodal queries.

Research Intern, *(Indian Institute of Science)* **Bengaluru, India** *05/2022 - 08/2022*

- Developed a source-free domain-adaptation framework for image classification under target-domain shift.

Undergraduate Research Assistant, *(Jadavpur University)* **Kolkata, India** *10/2020 - 05/2022*

- Graph learning and event-based vision for signal recovery, semi-supervised segmentation, and moving-object detection.

Education

Ph.D. Artificial Intelligence *University of Surrey (CVSSP & PAI)* **Guildford, UK** *2022 - present*
Research on unified vision–language models, generative AI, RL post-training, and object counting; first-author papers in ACM TOG, AAAI, ICCV-W; TA in 2 MSc courses; Research Intern at Adobe Research. Expected defense: late 2026.

B.E. (Hons.) Electronics & Telecom. Engineering *Jadavpur University* **Kolkata, India** *2018 - 2022*
GPA 8.79/10; research on graph signal processing and event-based vision; published at ICASSP and EUSIPCO; 2 years of undergraduate research across vision and signal recovery.

Selected Publications*

- **ABACUS: Adapting a Unified Foundation Model for Bridging Image Count Understanding and Generation** *ACM Trans. of Graphics (SIGGRAPH Asia)* 2026
  Anindya Mondal, Sauradip Nag, Anjan Dutta
  We build a unified foundation model that bridges image count understanding and generation, aligned with a cycle-consistent GRPO objective.
- **OmniCount: Multi-label Object Counting with Semantic–Geometric Priors** *AAAI* 2025
  Anindya Mondal, Sauradip Nag, Xiatian Zhu, Anjan Dutta
  We propose a training-free framework for open-vocabulary multi-label counting using semantic and geometric priors.
- **CountLoop: Training-Free High-Instance Image Generation via Iterative Agent Guidance** *Under Review* 2025
  Anindya Mondal, Ayan Banerjee, Sauradip Nag, Josep Lladós, Xiatian Zhu, Anjan Dutta
  We propose the first training-free high-instance image generation method using an MLLM planner–critic loop.
- **Actor-Agnostic Multi-label Action Recognition with Multi-modal Query (MSQNet)** *ICCV-W* 2023
  Anindya Mondal, Sauradip Nag, Joaquin M. Prada, Xiatian Zhu, Anjan Dutta
  We propose an actor-agnostic multi-label action recognition model with multimodal queries (+50% mAP on Animal Kingdom).
- **Time-Varying Signals Recovery via Graph Neural Networks** *ICASSP* 2023
  J. A. C. Correa, J. H. Giraldo, **Anindya Mondal**, et al.
  We recover missing time-varying graph signals using graph neural networks.
- **Recovery of Missing Sensor Data by Reconstructing Time-Varying Graph Signals** *EUSIPCO* 2022
  Anindya Mondal, Mayukhmali Das, Aditi Chatterjee, Palaniandavar Venkateswaran
  We recover missing wireless sensor data as smooth time-varying graph signals via Sobolev-norm minimization.
- **Moving Object Detection for Event-based Vision using Graph Spectral Clustering** *ICCV-W* 2021
  Anindya Mondal, Shashant R, Jhony H. Giraldo, Thierry Bouwmans, Ananda S. Chowdhury
  We detect moving objects from event cameras fully unsupervised via graph spectral clustering.

* **Bold** = candidate; full publication list on Google Scholar.

Teaching and Service

- **Teaching Assistant**, University of Surrey (2023–2025): Applied Machine Learning (EEEM068) and Advanced Topics in Computer Vision & Deep Learning (EEEM071); mentored MSc AI cohorts.
- **Reviewer:** CVPR, ICCV, ECCV, NeurIPS, BMVC, ICASSP, ICPR, Pattern Recognition, IEEE TSP, IEEE TSIPN.

Honors and Awards

- **AAAI 2025 Grant:** Conference and travel grant for OmniCount.
- **ICCV 2023 Grant:** Conference grant.
- **Surrey Postgraduate Studentship:** Full PhD funding.
- **ACM SIGKDD India Chapter:** Uplink Research Internship Award.